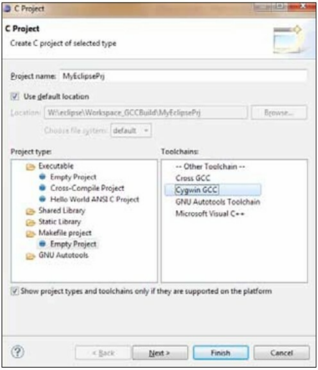

Figure 1: Makefile project

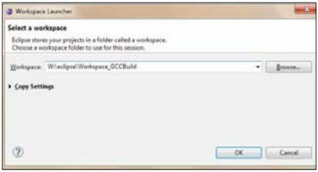

Figure 2: Project settings

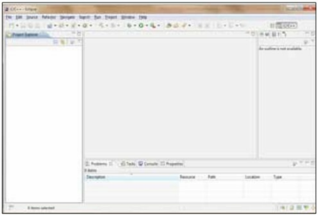

Figure 3: GCC source in Eclipse IDE

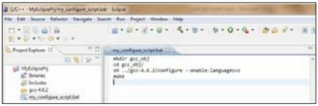

Figure 4: Configuration for your batch file

Step 4: Creating a project. Now, you need to add a project. Click *File->New->C Project*. Give a name to the project; it should be a *Makefile* project with the Cygwin GCC tool-chain, as shown in Figure 1. (I repeat, this option will be visible only if Cygwin is installed on your system.) This will open up an empty project.

Step 5: Project settings—make sure *Build Automatically* is selected in the *Project* menu, as shown in Figure 2.

Step 6: Using the file manager (Explorer), place the extracted GCC source folder (e.g., *gcc-4.6.2*) in the folder created for your project. Now, refresh the workspace in Eclipse by pressing F5. The source is displayed in the *Project Explorer*, as shown in Figure 3.

Step 7: Now, you need to see how to compile the source. This part is a bit tricky. For this, create a *.bat* batch file containing the *configure* command, as shown below:

```
mkdir gcc_obj
cd gcc_obj/
sh ../gcc-4.6.2/configure --enable-languages=c
make
```

Note: As mentioned earlier, I assume you're familiar with the GCC build process. Here, we are building GCC for native.

Add this *.bat* file to your project folder (at the same level as the GCC source folder), and refresh the workspace, before checking that it's shown.

Step 8: Next, register this batch file as an external tool. For that, go to *Run->External Tools->External Tools Configuration*. Now, specify a name for the external tool. Click *Browse Workspace* and select your batch file from the listing for the *Location* field. Specify your project directory as the working directory (see Figure 4). Now, hold your breath, and click *Run*. Voilà, your code has started compiling!

So, all it takes is 8 simple steps to build GCC source using the Eclipse IDE. It is even more interesting to debug your GCC code using Eclipse. Watch out for my next article to learn more about that! END

By: Deepti Sharma

The author is an open source enthusiast with almost a decade of experience in design, development and quality validation of systems software tools (compilers, assemblers and device drivers) for embedded systems. Currently working as a project manager (Software Group) at RF-Silicon Pvt Ltd, she has also been mentor for training on GCC and compiler development, following good practices and establishing processes for maintaining quality. She loves writing for LFY. She can be reached at *deepti.acmet@gmail.com*.